

Homework 11

Arham Lodha

April 11, 2024

Problem 1.

Proof. Suppose f is continuous at 0 with $f(0) > 0$. Let $\epsilon = f(0)/2$. By the definition of continuity, for this ϵ , $\exists \delta > 0$ such that $\forall x \in \mathbb{R} \ni |x - 0| = |x| < \delta$, $-\epsilon < f(x) - f(0) < \epsilon$. Thus, $-\epsilon + f(0) = \frac{f(0)}{2} < f(x) < \frac{3f(0)}{2}$. $\frac{3f(0)}{2} - 2f(0) = f(0)(\frac{3}{2} - 2) = -\frac{f(0)}{2} < 0$, since $f(0) > 0$. Thus $\frac{3f(0)}{2} < 2f(0)$. Thus $\frac{f(0)}{2} < f(x) < \frac{3f(0)}{2} < 2f(0)$ and thus $\frac{f(0)}{2} < f(x) < 2f(0)$. Thus, $\frac{f(0)}{2} \leq f(x) \leq 2f(0)$. $\square$

Problem 2a.

Proof. $f : \mathbb{R} \rightarrow \mathbb{R}$ where $\forall x \in \mathbb{R}$:

$$f(x) := \begin{cases} x & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \in \mathbb{R} \setminus \mathbb{Q} \end{cases}$$

Thus f is defined on $x = 0$.

$\forall \epsilon > 0$, let $\delta = \epsilon$. $\forall x \in \mathbb{R} \ni |x - 0| = |x| < \delta$:

1. If, $x \in \mathbb{Q}$:

$$\begin{aligned} |f(x) - f(0)| &= |x - 0| \\ &< \delta = \epsilon \end{aligned}$$

2. If, $x \in \mathbb{R} \setminus \mathbb{Q}$:

$$|f(x) - f(0)| = 0 < \epsilon$$

Thus, $|f(x) - f(0)| < \epsilon$. Thus by definition of continuity, f is continuous at $c = 0$. $\square$

Problem 2b.

Proof. $\forall c \in \mathbb{R} \ni c \neq 0$.

Let

$$x_n = \frac{\lfloor c * 10^n \rfloor}{10^n}$$

and

$$y_n = c + \frac{1}{\pi^n}$$

Both x_n and y_n converge to c . Furthermore x_n is a rational sequence because 10^n is a integer and $\lfloor c * 10^n \rfloor$ is a integer as well. y_n is a irrational sequence because π is irrational, so π^n is irrational so $\frac{1}{\pi^n}$ is irrational and finally $c + \frac{1}{\pi^n}$ is irrational.
 $\forall \epsilon > 0$, let $K = 1$. $\forall n \geq K$,

$$|f(y_n) - 0| = |0 - 0| = 0 < \epsilon$$

and

Since x_n converges to c , for this ϵ , $\exists M \in \mathbb{N} \ni \forall n \geq M$, $|x_n - c| < \epsilon$.

$$|f(x_n) - x| = |x_n - x| < \epsilon$$

Thus $\lim_{n \rightarrow \infty} f(y_n) = 0$ and $\lim_{n \rightarrow \infty} f(x_n) = 0$. However, $\lim_{n \rightarrow \infty} f(x_n) = c \neq 0 = \lim_{n \rightarrow \infty} f(y_n)$.
 By the sequential criterion for discontinuity, f is discontinuous. $\square$

Problem 3.

Proof. Let $A \subset \mathbb{R}$. Suppose $f : A \rightarrow \mathbb{R}$ where f is unbounded on A . Construct the sequence (x_n) in A as follows. For each $n \in \mathbb{N}$, by the definition of unbounded, $\exists x \in A \ni f(x) > n$, thus let $x_n = x$.
 $\forall \alpha \in \mathbb{R}$.

1. If $\alpha \leq 0$, let $K = 1$ thus $\alpha < K$. $\forall n \geq K$, since $f(x_n) > n \geq K > \alpha$, $f(x_n) > \alpha$.
2. If $\alpha > 0$, by the Archemedian property, $\exists K \in \mathbb{N} \ni K > \alpha$. $\forall n \geq K$, since $f(x_n) > n \geq K > \alpha \implies f(x_n) > \alpha$.

Thus $(f(x_n))$ is properly divergent to infinity and $\lim_{n \rightarrow \infty} f(x_n) = +\infty$. $\square$